SECTION 2: PROJECT PLAN

Banking Transaction System – Requirements & Design Specification

2. Project Plan

2.1 Work Breakdown Structure (WBS)

The project scope is deconstructed into six core execution phases aligned with the system modules:

- **Phase 1: Project Setup & Core Domain (AUTH, CUS)**
  - Initialize Git repository, Spring Boot 3.5.x skeleton, Maven configuration, and dev seed data.
  - Validate SQLite connection pooling, WAL mode, and single-writer immediate transaction architecture spike.
  - Build domain model classes and implement registration, login, session handling, and role-based access control (CUSTOMER, ADMIN).
- **Phase 2: Core Banking Operations & Ledger (ACC, TXN)**
  - Build domain model classes for Savings, Checking, and Fixed Deposit account types.
  - Implement deposit, withdrawal, and transfer service methods with atomic balance updates.
  - Enforce transaction limits, account withdrawal rules, dynamic transfer fee calculations, and append-only ledger entries.
- **Phase 3: Reporting, Administration, & Scheduling (STM, ADM, JOB, AUD)**
  - Create account statement views with pagination, date filtering, and CSV export functionality.
  - Build admin dashboard metrics, customer search, account freeze/unfreeze endpoints, and audit logging.
  - Implement scheduled jobs for monthly savings interest accrual and fixed deposit maturity processing, and freeze HTML element IDs.
- **Phase 4: Automated & Security Testing**
  - Develop unit tests for domain rules and MockMvc integration tests for REST controllers.
  - Build Postman API test collection and Selenium 4 WebDriver UI test scripts using stable element IDs.
  - Execute security testing targeting OWASP Top 10 risks (IDOR, privilege escalation, XSS, CSRF).
- **Phase 5: Quality Assurance & Defect Remediation**
  - Conduct static code analysis with SonarLint / SpotBugs and perform OWASP dependency vulnerability checks.
  - Resolve bugs logged in Jira and generate JaCoCo code coverage reports.
- **Phase 6: Documentation, Final Rehearsal & Delivery**
  - Generate OpenAPI/Swagger documentation and finalize Jira test-activity reports.
  - Conduct system demo rehearsals, final code defense, and project presentation.

2.2 Timeline & Milestones

The compressed 6-week schedule defines key delivery dates, dependencies, and project milestones:

- **Schedule Overview:**
  - **Week 1:** Requirements signoff, repository setup, database schema, SQLite spike validation, domain model, registration, login, and roles.
  - **Week 2:** Accounts, deposits, withdrawals, transfers, REST API endpoints, UI forms, error handling, dynamic fee calculations, and ledger entries.

- **Week 3:** Statements, CSV export, admin dashboard, audit log, scheduled jobs, and freeze HTML element IDs for test automation.
- **Week 4:** Selenium UI automation, Postman API collection, and security test suite execution.
- **Week 5:** Static testing (SonarLint/SpotBugs), bug fixing, full regression testing, and JaCoCo coverage report.
- **Week 6:** OpenAPI/Swagger documentation, Jira test-activity reports, demo rehearsal, and final project presentation/defense.

- **Key Project Milestones:**
  - **Milestone 1 (Week 1):** Project Foundations & Auth Signoff (Database schema, seed data, user auth, and SQLite spike verified).
  - **Milestone 2 (Week 2):** Core Banking MVP (Complete transaction engine with fee calculation and ledger logging).
  - **Milestone 3 (Week 3):** Feature Complete & UI Freeze (Admin tools, statements, frozen DOM IDs for automation).
  - **Milestone 4 (Week 5):** Code Quality & Test Signoff (Automated tests passing, static analysis completed, zero critical bugs open).
  - **Milestone 5 (Week 6):** Final System Delivery & Presentation.

2.3 Resource Plan

The required technology stack, developer tools, hardware, and runtime environments are detailed below:

Resource Category	Tools & Environment Details
Backend Framework	Java 17 or higher, Spring Boot 3.5.x (Web, Data JPA, Security, Validation), Maven
Database & Storage	SQLite (embedded single-file database), sqlite-jdbc, Hibernate SQLite dialect, DB Browser for SQLite
Frontend & UI	Thymeleaf view templates, plain CSS (or Bootstrap framework)
Testing Tools	JUnit 5, Mockito, Spring Boot Test / MockMvc, Selenium 4 (Page Object Model), Postman, JaCoCo
Quality & Process	Git + GitHub (pull requests), Jira (board, bug tracking), SonarLint / SpotBugs, OWASP Dependency-Check
Hardware / Host	Standard developer workstations running JDK 17+ and local embedded database environment